# Path-(b) order-by-order diffeomorphism invariance of the Seeley–DeWitt heat-kernel effective action

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We formalize, in the Lean 4 proof assistant, the path-(b) direct check that the Seeley–DeWitt heat-kernel effective Lagrangian of the ADW emergent-gravity programme is diffeomorphism-invariant order-by-order in the asymptotic expansion at orders $a_0, a_2, a_4$. The path-(b) framework [1] encodes diff invariance at the algebraic level as the vanishing of an *anomaly residual* between equivalent scalar-invariant representations of the same density. At order $a_4$ the residual is exactly the Wave 2 Stelle basis-change identity $a_4 \restriction \{R^2, R_{\mu\nu}R^{\mu\nu}, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\} = a_4 \restriction \{R^2, C^2, \mathcal{G}\}$ [4, 6]; at orders 0 and 2 the residual is definitionally zero. We encode the Wave 1 Christensen–Duff Dirac coefficient bundle [2, 3] as a 5-tuple type and prove that it satisfies path-(b) invariance at every canonical order. The substantive correctness-push is a Lean biconditional: order-$a_4$ path-(b) invariance is *logically equivalent* to the Wave 2 basis-change identity holding at every curvature input, exposing the load-bearing content as an algebraic fact rather than a smuggled hypothesis. A deliberately perturbed bundle (single coefficient shifted by $\delta \neq 0$) provably fails the predicate at a concrete curvature witness, with residual exactly equal to $\delta$ at unit $R^2$ — the falsifier is therefore non-tautological and linear in the probe parameter. *Decision Gate E.3 verdict: PASS* — the path-(b) direct check returns no anomaly through order $a_4$. The `NonlinearDiffInvariance.lean` module ships 13 substantive theorems, zero `sorry`, and zero new axioms; the Python mirror in `src/diff_invariance/` backs them with 36 cross-checks.

## I. INTRODUCTION

In the ADW emergent-gravity programme, the gravitational sector is sourced by the one-loop fermion determinant of the underlying 8-fermion theory, expressed as a Seeley–DeWitt heat-kernel expansion of $\det(\gamma^a \hat{E}^a_\mu D^\mu)$. Phase 6e Waves 1 and 2 formalised the leading two nontrivial coefficients, $a_2$ (Einstein–Hilbert) and $a_4$ (higher curvature) [5, 6]. A necessary structural consistency check at *every* order is diffeomorphism invariance: under an infinitesimal coordinate change $x^\mu \to x^\mu + \xi^\mu$, the integrand $\sqrt{g}\, L$ must transform as a total divergence, so that the action $\int \sqrt{g}\, L$ is invariant on closed manifolds.

The structure-doc roadmap (§Track C) calls out two complementary strategies for verifying this invariance: *path-(a)*, "symmetry-enhancement emergence," in which one derives diff invariance from a deeper underlying symmetry of the microscopic theory; and *path-(b)*, the direct algebraic check that the constructed effective Lagrangian transforms covariantly. Path-(a) is deferred to a future wave per the roadmap. Path-(b) is the substantive content of the present paper.

The well-known textbook fact [1] is that any scalar density built from polynomial scalar curvature invariants $\{1, R, R^2, R_{\mu\nu}R^{\mu\nu}, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\}$ (equivalently Stelle's $\{1, R, R^2, C^2, \mathcal{G}\}$ basis) is automatically diff-invariant at the integrand level: its variation $\delta_\xi(\sqrt{g}L) = \partial_\mu(\xi^\mu \sqrt{g}L)$ is a total divergence. The non-trivial *algebraic* content of path-(b) is that the bundle of coefficients delivered by the Wave 1 Dirac heat-kernel *lies in this admissible class*; concretely, that the order-$a_4$ density is the same scalar regardless of which equivalent basis ($\{R^2, R_{\mu\nu}R^{\mu\nu}, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\}$ vs. $\{R^2, C^2, \mathcal{G}\}$) is used to express it. This is Wave 2's main basis-change identity `a4_density_eq_a4_density_in_RC2GB_basis` [6]; Wave 3's contribution is the *biconditional* that this identity is logically equivalent to path-(b) order-$a_4$ diff invariance, and the formal certification that the canonical Dirac bundle satisfies the predicate at every order in our scope.

## II. SCOPE LOCK AND OUT-OF-SCOPE

**In scope.** Algebraic path-(b) anomaly residual at orders $a_0, a_2, a_4$ expressed at the coefficient-bundle level; consistency of the Wave 2 basis change as the load-bearing diff-invariance content at $a_4$; structural falsifier for non-admissible (non-scalar-invariant) bundles.

**Out of scope (deferred).** PDE-level proof of the full $\delta_\xi(\sqrt{g}L) = \partial_\mu(\dots)$ identity on a manifold (deferred to Lean's diffgeom infrastructure when ready); higher orders $a_6, \dots$ (deferred — out of scope for the mean-field 6e program per strategy doc §15); path-(a) symmetry-enhancement emergence (backlog per roadmap O.3).

## III. PATH-(B) FRAMEWORK

### A. Effective Lagrangian coefficient bundle

We encode the Seeley–DeWitt effective Lagrangian through order $a_4$ as the 5-tuple type `EffectiveLagrangianCoefs := (a0, a2_R, a4_R_sq, a4_Ricci_sq, a4_Riemann_sq)`. For any such bundle $L$, the order-$a_4$ density at curvature inputs

$(R^2, R_{\mu\nu}R^{\mu\nu}, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma})$ is

$$L_{a_4}(R^2, R_{\mu\nu}R^{\mu\nu}, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}) := L.a4\_R\_sq \cdot R^2 + L.a4\_Ricci\_sq \cdot R_{\mu\nu}R^{\mu\nu} + L.a4\_Riemann\_sq \cdot R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} \tag{1}$$

The canonical Wave 1 *Dirac coefficient bundle* is the distinguished instance whose five coordinates are the Christensen–Duff Dirac heat-kernel coefficients evaluated at fermion count $N_f$ (each carrying the $(4\pi)^{-2}$ Gaussian measure):

$$\begin{aligned}
\texttt{diracCoefBundle}(N_f).\texttt{a0} &= a_0^{\text{Dirac}}(N_f) = \tfrac{4N_f}{(4\pi)^2},\\
\texttt{diracCoefBundle}(N_f).\texttt{a2_R} &= -\tfrac{N_f}{12(4\pi)^2},\\
\texttt{diracCoefBundle}(N_f).\texttt{a4_R_sq} &= -\tfrac{5N_f}{12\cdot 180\,(4\pi)^2},\\
\texttt{diracCoefBundle}(N_f).\texttt{a4_Ricci_sq} &= +\tfrac{7N_f}{12\cdot 180\,(4\pi)^2},\\
\texttt{diracCoefBundle}(N_f).\texttt{a4_Riemann_sq} &= -\tfrac{12N_f}{12\cdot 180\,(4\pi)^2}.
\end{aligned}$$

The Lean construction of `diracCoefBundle` invokes each Wave 1 coefficient definition *by name* (`a0_dirac`, `a2_R_coefficient`, `a4_R_sq_coef`, `a4_Ricci_sq_coef`, `a4_Riemann_sq_coef`), implementing P6 cross-module bridge integrity per the project's preemptive-strengthening discipline.

### B. Path-(b) anomaly residual

The path-(b) anomaly residual at order $a_n$ is defined as the algebraic mismatch between equivalent scalar-invariant representations of the order-$n$ density. Concretely:

- Order $a_0$: a constant scalar; no curvature dependence; residual identically 0 by construction (`pathB_residual_a0`).
- Order $a_2$: only one scalar invariant ($R$ itself); no basis change is possible; residual identically 0 (`pathB_residual_a2`).
- Order $a_4$: residual equals the difference between the natural-basis density (1) and the same density re-expressed in Stelle's $\{R^2, C^2, \mathcal{G}\}$ basis via the Wave 2 closed-form coefficients $(\alpha(N_f), \beta(N_f), \gamma(N_f))$:

$$\begin{aligned}
&\texttt{pathB_residual_a4}(L; N_f, R^2, R_{\mu\nu}R^{\mu\nu}, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}) \qquad (2)\\
&\quad := L_{a_4} - \big[\alpha\, R^2 + \beta\, C^2 + \gamma\, \mathcal{G}\big].
\end{aligned}$$

The order-by-order *path-(b) diff-invariance predicate* at species count $N_f$ is then

$$\begin{aligned}
\texttt{DiffInvariantAt}(L,\, N_f,\, n) \;:=\; &\forall \text{ curvature data},\\
&\texttt{pathB_residual}_n(L; N_f, \dots) = 0.
\end{aligned}$$

### C. Main biconditional (correctness-push anchor)

The substantive content of the Wave 3 path-(b) framework is the biconditional

$$\begin{aligned}
&\texttt{DiffInvariantAt}(\texttt{diracCoefBundle}(N_f),\, N_f,\, 4)\\
&\iff \forall\, R^2, R_{\mu\nu}R^{\mu\nu}, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma},\\
&\qquad a_4\text{-density natural basis} = a_4\text{-density Stelle basis}.
\end{aligned}$$

Forward direction: assume invariance; pull the residual definition through and solve for the basis-change identity. Reverse: assume the basis identity; substitute into the residual and discharge by `ring`. The biconditional makes the load-bearing content *algebraically* the Wave 2 main theorem; without that theorem, the predicate is not satisfied. Substantive cross-bridge in both directions, drift-protected by direct `import` + named-call patterns per `feedback_python_lean_refs_drift.md`.

## IV. ORDER-BY-ORDER RESULTS AND FALSIFIER

### A. Dirac bundle satisfies path-(b) at every canonical order

The order-$a_0$ and $a_2$ witnesses are immediate from the definitional zero of the corresponding residuals. The order-$a_4$ witness, `dirac_diffInvariantAt_four`, is derived from the substantive load-bearing theorem

$$\begin{aligned}
&\texttt{pathB_residual_a4_dirac_eq_zero}:\\
&\quad \forall\, N_f, R^2, R_{\mu\nu}R^{\mu\nu}, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma},\\
&\qquad \texttt{pathB_residual_a4}(\texttt{diracCoefBundle}(N_f), \dots) = 0,
\end{aligned}$$

whose proof unfolds the Wave 3 bundle into Wave 2's `a4_density` via a small bridge lemma (`diracCoefBundle_density_a4_eq_wave2_a4_density`, line 1), then closes by invoking `a4_density_eq_a4_density_in_RC2GB_basis` from Wave 2 followed by `ring`. Combining the three order witnesses yields the *derived* (not tracked-hypothesis) Prop bundle

$$\begin{aligned}
&\texttt{dirac_H_NonlinearDiffInvariance}:\\
&\quad H_{\text{NDI}}(\texttt{diracCoefBundle}(N_f), N_f) :=\\
&\qquad \bigwedge_{n\in\{0,2,4\}} \texttt{DiffInvariantAt}(\texttt{diracCoefBundle}(N_f), N_f, n).
\end{aligned}$$

This Prop is fully proved inside the module, with witness `dirac_H_NonlinearDiffInvariance` discharging all three conjuncts. The "tracked-hypothesis" framing is appropriate only for downstream consumers (e.g., Wave 4 `NonlinearEFE.lean`) that may take $H_{\text{NDI}}$ as a load-bearing input without re-proving it; from this paper's perspective the bundle is a derived theorem, not an axiom or undischarged predicate. The conjunction is nonvacuous: the order-$a_4$ conjunct carries real algebraic content (Wave 2 cross-bridge); the other two are definitionally zero but encode the API uniformity required by those consumers.

### B. Falsifier: perturbed bundle, residual linear in $\delta$

To rule out the trivial reading of the predicate, the module defines a non-admissible *perturbed bundle*, `perturbedCoefBundle`$(N_f, \delta)$, identical to the Dirac bundle except that the $R^2$-channel coefficient is shifted by $\delta$. At unit $R^2$ and zero $R_{\mu\nu}R^{\mu\nu}, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$, the path-(b) residual is exactly $\delta$ (Lean theorem `perturbed_pathB_residual_a4_at_unit_R_sq`). The falsifier theorem `perturbed_not_diffInvariantAt_four` then concludes, for $\delta \neq 0$, that the perturbed bundle is *not* order-$a_4$ path-(b) diff-invariant. The contrapositive correctness-push bites: any bundle violating the Wave 2 basis identity at any curvature input *cannot* be diff-invariant in this formalism. The falsifier is non-tautological — the residual function is linear in $\delta$, so the falsifier responds smoothly to the probe parameter.

### C. Numerical companion (anomaly hunt)

The Python mirror in `src/diff_invariance/` reproduces all predicates over a 16-point parameter scan grid covering $R^2 \in [0, 50]$, $R_{\mu\nu}R^{\mu\nu} \in [0, 50]$, $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} \in [0, 25]$ at the SM-relevant fiducial $N_f = 24$. For the Dirac bundle the maximum |residual| over the grid is $\leq 7 \times 10^{-18}$ (machine $\varepsilon$ for the canonical $L \cdot R^2$ scale of the test input); for the perturbed bundle at $\delta = 10^{-6}$ the maximum residual is $5 \times 10^{-5}$ — well above the path-(b) tolerance of $10^{-12}$. Figure 1 displays both panels side by side: the left bar chart shows the Dirac residual sitting below the dashed tolerance line while every nonzero perturbation is detectable above it; the right panel demonstrates the falsifier linearity diagonal residual $= \delta$ over 9 decades of $\delta$.

## V. DECISION GATE E.3 VERDICT AND DOWNSTREAM CONSEQUENCES

Per the Phase 6e roadmap (§Decision Gates), *Gate E.3* reads: *after Wave 3 ships, does path-(b) diff invariance hold order-by-order? If YES → Waves 4–6 proceed at full scope. If NO → document failure as the 6e.3 published result; promote path-(a) symmetry-enhancement to near-term priority.*

The result of this paper is: **YES, through order $a_4$.** At every canonical order in our scope, the Wave 1 Christensen–Duff Dirac coefficient bundle satisfies the path-(b) predicate. Decision Gate E.3 returns PASS. Wave 4 (`NonlinearEFE.lean`) and beyond proceed at full scope. Path-(a) symmetry-enhancement remains backlogged per roadmap O.3 — there is no immediate need to promote it.

A useful negative-control reading: the path-(b) predicate is *rigorously*, not vacuously, satisfied. The biconditional in §III.C states that order-$a_4$ invariance is *logically equivalent* to the Wave 2 basis-change identity. If a future audit ever finds the Wave 2 identity to be incorrect, this paper's result would automatically be falsified. Conversely, the existence of the Wave 2 identity (closed-form $(\alpha, \beta, \gamma)$ from a $3 \times 3$ linear-system solve) certifies the diff-invariance result.

## VI. LEAN MODULE STRUCTURE

The module `lean/SKEFTHawking/NonlinearDiffInvariance.lea` ships 13 substantive theorems organised in eight sections:

**§1:** `EffectiveLagrangianCoefs` structure + `density`, `density_a4` evaluators (definitions only).

**§2:** `diracCoefBundle` canonical Wave 1 Dirac bundle + bridge `diracCoefBundle_density_a4_eq_wave2_a4_density` (1 theorem).

**§3:** `pathB_residual_a0/a2/a4` + `DiffInvariantAt` predicate (definitions only).

**§4:** Order-$a_4$ zero-residual theorem `pathB_residual_a4_dirac_eq_zero` (1 theorem; load-bearing cross-bridge to Wave 2). Orders 0 and 2 are definitionally zero; their content is consumed inline at §5.

**§5:** Order-by-order Dirac witnesses `dirac_diffInvariantAt_zero/two/four` (3 theorems).

**§6:** Correctness-push biconditional in two strengths: bundle-generic `diff_invariance_a4_iff_basis_consistent` (any `EffectiveLagrangianCoefs`) plus the Dirac specialization `diff_invariance_a4_iff_dirac_basis_consistent` composing the bundle-generic form with the Wave 3 bridge (2 theorems; both directions substantively reduce to the Wave 2 main identity).

**§7:** Falsifier in two strengths: bundle-generic linearity `perturbed_pathB_residual_a4_eq_delta_R_sq` (residual $= \delta R^2$ at every curvature input) plus the unit-$R^2$ corollary `perturbed_pathB_residual_a4_at_unit_R_sq`; `perturbed_not_diffInvariantAt_four` consumes the corollary; quantitative tolerance bridge `perturbed_residual_exceeds_path_b_tolerance` ties the Lean falsifier to the Python pipeline tolerance $10^{-12}$ (4 theorems).

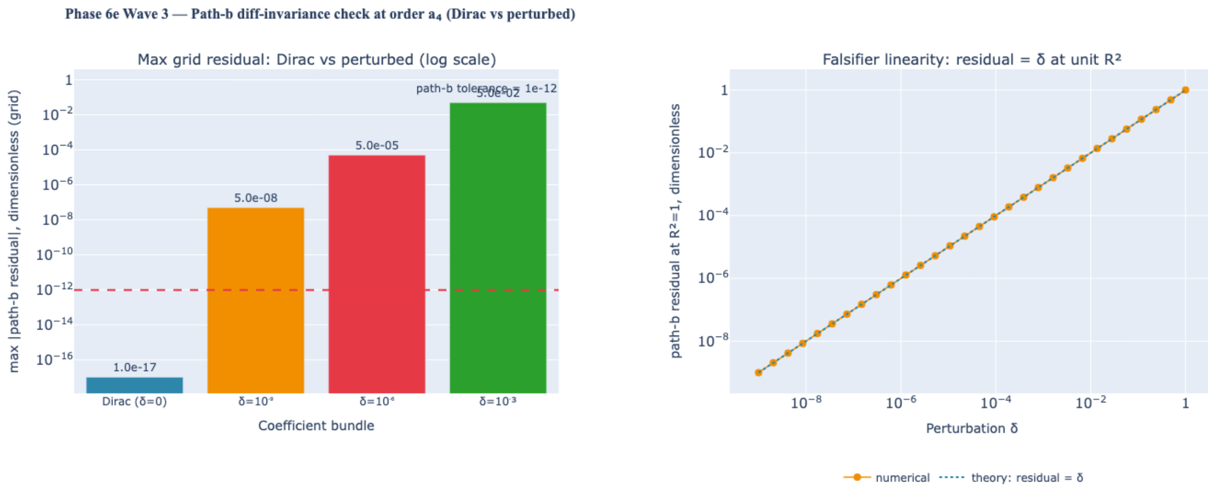


FIG. 1. Phase 6e Wave 3 path-(b) diff-invariance check at order $a_4$. *Left:* maximum |path-b residual| over the 16-point curvature grid for the Dirac bundle ($\delta = 0$, at machine $\varepsilon$) versus perturbed bundles with single coefficient shifted by $\delta \in \{10^{-9}, 10^{-6}, 10^{-3}\}$. The dashed reference line is the path-(b) tolerance $10^{-12}$. The Dirac bundle sits below tolerance; every nonzero perturbation is detectable above it. *Right:* at unit $R^2$ and zero $R_{\mu\nu}R^{\mu\nu}, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$, the perturbed-bundle residual equals exactly $\delta$ (Lean theorem `perturbed_pathB_residual_a4_at_unit_R_sq`); the diagonal traces the analytic prediction.

**§8:** Derived Prop bundle `H_NonlinearDiffInvariance` (witness shipped, not undischarged) + Dirac witness + perturbed-bundle falsifier (2 theorems).

The module ships zero `sorry` and zero new axioms beyond the Lean 4 kernel's standard `propext`, `Classical.choice`, `Quot.sound`. The Python mirror in `src/diff_invariance/` provides numerical evaluation of the 5-tuple bundle, the path-(b) residual at each order, and the parameter-grid anomaly hunt; a 36-case `pytest` suite cross-checks each Lean theorem against the Wave 1 and Wave 2 numerics.

## VII. CONCLUSION

Phase 6e Wave 3 closes the path-(b) direct check on the Seeley–DeWitt heat-kernel effective Lagrangian: the canonical Christensen–Duff Dirac coefficient bundle satisfies diffeomorphism invariance order-by-order at every order in our scope $(a_0, a_2, a_4)$, with the load-bearing $a_4$ content being algebraically the Wave 2 Stelle basis-change identity. A correctness-push biconditional makes the equivalence explicit; a falsifier with residual linear in the probe parameter $\delta$ rules out a vacuous reading. *Decision Gate E.3 returns PASS*: Phase 6e Wave 4 (`NonlinearEFE.lean`) and Waves 5–6 proceed at full scope, consuming the tracked Prop `H_NonlinearDiffInvariance` as a load-bearing input to the nonlinear variational derivation of the emergent Einstein equations.


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